DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY LUCKNOW

Evaluation Scheme & Syllabus

For

B.Tech. 2nd Year

(Food Technology)

On

AICTE MODEL CURRICULUM

(Effective from the Session: 2019-20)

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY LUCKNOW

B.TECH (Food Technology)

SEMESTER- III

Sl. No.	Subject Codes	Subject	Periods			Evaluation Scheme				End Semester		Total	Credit
			L	T	P	CT	TA	Total	PS	TE	PE		
1	KOE031-38/ KAS302	Engineering Science Course/Maths IV	3	1	0	30	20	50		100		150	4
2	KAS301/ KVE 301	Technical Communication/Universal Human values	2 3	1 0	0 0	30	20	50		100		150	3
3	KCH301	Material and Energy Balance	3	1	0	30	20	50		100		150	4
4	KFT301	Composition, Quality & Safety of Foods	3	1	0	30	20	50		100		150	4
5	KFT302	Food Microbiology	3	0	0	30	20	50		100		150	3
6	KFT351	Analysis of Food Constituents Lab	0	0	2				25		25	50	1
7	KFT352	Food Microbiology Lab	0	0	2				25		25	50	1
8	KFT353	Fluid Flow & Solid Handling Lab	0	0	2				25		25	50	1
9	KFT354	Mini Project or Internship Assessment*	0	0	2			50				50	1
10	KNC301/ KNC302	Computer System Security/Python Programming	2	0	0	15	10	25		50			0
11		MOOCs (Essential for Hons. Degree)											
		Total										**950**	**22**

*The Mini Project or internship (3-4 weeks) conducted during summer break after II semester and will be assessed during III semester.

SEMESTER- IV													
Sl. No.	**Subject Codes**	**Subject**	**Periods**			**Evaluation Scheme**				**End Semester**		**Total**	**Credit**
			L	**T**	**P**	**CT**	**TA**	**Total**	**PS**	**TE**	**PE**		
1	KAS402/ KOE041-48	Maths IV/Engineering Science Course	3	1	0	30	20	50		100		150	4
2	KVE401/ KAS401	Universal Human Values/ Technical Communication	3 2	0 1	0 0	30	20	50		100		150	3
3	KFT401	Heat & Mass Transfer	3	0	0	30	20	50		100		150	3
4	KFT402	Food Chemistry	3	1	0	30	20	50		100		150	4
5	KFT403	Food Biochemistry	3	1	0	30	20	50		100		150	4
6	KFT451	Heat & Mass Transfer Lab	0	0	2				25		25	50	1
7	KFT452	Food Chemistry Lab	0	0	2				25		25	50	1
8	KFT453	Food Biochemistry Lab	0	0	2				25		25	50	1
9	KNC402/ KNC401	Python Programming/ Computer System Security	2	0	0	15	10	25		50			0
10		MOOCs (Essential for Hons. Degree)											
		Total										**900**	**21**

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY

B.TECH II YEAR III SEMESTER FOOD TECHNOLOGY COURSE

SUBJECT CODE:**KCH 301**	COURSE TITLE: **Material And Energy Balance**
EXAMINATION DURATION: **3 Hrs.**	SEMESTER: **III (ODD)**
L:3 T:1 P:0 **CREDITS:04**	**PRE-REQUISITE:** NIL

OBJECTIVE: To provide basic knowledge of principles of material and energy balances applied to chemical engineering systems.

COURSE OUTCOME:

After successful completion of the course the students will be able to:

- Apply steady-state and unsteady state material and energy balance on a system.
- Analyze all the stiochiometric and balances being applied on a system undergoing Chemical Process
- Design equipment with inlet and outlet; including recycle- bypass streams for a chemical process .

REFERENCE BOOKS:

S. No	Name of Authors / Books / Publishers	Year of Publication/ Reprint
1.	Himmelblau D.M. and Riggs J. B.," Principles and Calculations in Chemical Engineering", 8th Ed., Prentice Hall of India.	2012
2.	Felder R.M. and Rousseau R.W., "Elementary Principles of Chemical Processes", 3rd Ed.., John Wiley.	2005
3.	Bhatt B.I. and Vora S.M., "Stoichiometry", 5th Ed., Tata McGraw-Hill	2010
4.	Narayanan K.V. and Lakshmikutty B., "Stoichiometry and Process Calculations", Prentice Hall of India.	2006
5.	Hougen D.A., Watson K.M. and Ragatz R.A., "Chemical Process Principles", Part-I, 2nd Ed., CBS Publishers.	1995

COURSE DETAILS:

Units	S. No.	Contents	Lecture Hours
I	**1.**	**Introduction:** Units and dimension in chemical engineering**,** units conversion of dimensional equations, stoichiometric and composition relations, concept of degrees of freedom and linear independence of a set of equations.	**5**
	2.	**Material Balance:** Concept of material balance, open and closed systems, steady state and unsteady state, multiple component system, selection of a basis, problem solving strategy.	**4**
II	**3.**	**Material Balance without Chemical Reaction for Single and Multiple Units:** Conservation of mass/atom, material balance for Systems without chemical reactions involving single unit and multiple units.	**5**
	4.	**Material Balance with Chemical Reaction for Single and Multiple Units:** Concept of excess reactant, extent of reaction, Material balance for systems with chemical reactions involving single unit and multiple units.	**6**
III	**5.**	**Recycle, Bypass, Purge and Industrial Applications:** Calculations for a cyclic processes involving recycle/ purge/ bypass, material balances involving gases, vapors, liquids and solids and use of real gas relationships, material balance involving gases, vapors, liquids & solids and uses of real gas relationships, vapor-liquid equilibrium and concepts of humidity & saturation, analysis of systems with bypass, recycle and purge, analysis of processes involving condensation, crystallization and vaporization.	**7**
IV	**6.**	**Energy Balance:** Conservation of energy with reference to general energy balance with and without chemical reactions, chemical engineering problems involving reversible processes and mechanical energy balance.	**4**
	7.	**Applications of Energy Balance:** Calculations of heat of change of phase (solid – liquid & liqid – vapor), heat of reaction, heat of combustion, heat of solutions and mixing, determination of temperatures for adiabatic and non-adiabatic reactions, use of psychometric and enthalpy- concentration diagrams.	**6**
V	**8.**	**Simultaneous Material and Energy Balances:** Degrees of freedom analysis for multicomponent systems, combined steady state material and energy balances for units with multiple sub-systems.	**3**
	9.	**Unsteady State Material and Energy Balances:** Transient materials and energy balances involving with and without chemical reactions.	**2**
		TOTAL	**42**

B.TECH II YEAR III SEMESTER FOOD TECHNOLOGY COURSE

SUBJECT CODE:**KFT 301**	COURSE TITLE: **COMPOSITION, QUALITY & SAFETY OF FOODS**
EXAMINATION DURATION: **3 Hrs.**	SEMESTER: **III (ODD)**
L: 3 T:1 P:0 CREDITS:04	**PRE-REQUISITE:** NIL

OBJECTIVE: To provide basic knowledge of Composition and Nutritional value of foods, Quality of Foods the relevant parameters and information about Safety of foods.

COURSE OUTCOME:

After successful completion of the course the students will be able to:

- Characterize the different categories of foods and their components.
- Understand the Nutritional value of foods, its estimation and Balanced Diet,
- Gain insight into the Quality aspects of foods and the relevant parameters and their estimation.
- Gain insight into Safety of foods, differentiate between Safe and Unsafe foods and know the hazards and risks associated with Unsafe foods.

REFERENCE BOOKS:

S. No	Name of Authors / Books / Publishers	Year of Publication/ Reprint
1.	Fennema O.R , "Food Chemistry" , 5TH Edition, Marcel Dekker	2017
2.	**Mever, Lillian Hoagland, " Food Chemistry",** CBS Publishers & Distributors Pvt. Ltd.,	2004
3.	Kramer, A, Twigg, B A (Eds), "Quality Control for the Food Industry (Volume I & II)", The AVI Publishing Company	1970
4.	Norman N.Potter, Joseph H.Hotchkiss, " Food Science", 5th Edition, CBS Publishers & Distributors Pvt. Ltd.,	2007

COURSE DETAILS:

Units	S. No.	Contents	Lecture Hours
I	1.	Composition: Chemical constituents of foods: Desirable and Potentially undesirable food constituents and their importance.	3
	2	Recommended Dietary Allowances (RDA). Basal metabolic rate and dietary requirements of different age groups.	3
	3	Composition of foods – General and specific for different foods of plant and animal origin.	2
II	1	Carbohydrates: Classes, Nomenclature and structure. Dietary utilization and disturbances	3
	2	Lipids: Definition, Classification and structure: Fatty acids composition of natural lipids of plants and animal origin	3
	3	Essential fatty acids. Role and use of natural lipids and tailor made fats in foods.	3
III	1	Protein: Physico-chemical properties of amino acids, peptides and proteins, structure - function relationship of proteins	5
	2	Essential Amino acids. Nutritional attributes of food and their implications	3
IV	1	Quality: Basic concepts. Nutritional and sensory attributes and their assessment	4
	2	Causes of undesirable changes leading to quality deterioration in foods and their implications.	3
	3	Determination of probable cause(s) of observed quality change in foods.	2
V	1	Safety: Operational sense of food safety, Potential Food derived health hazards- Microbial contamination	3
	2	Pesticide residues, Environmental Contamination. Risk and risk assessment.	2
	3	HACCP. Adulteration in Foods. Testing food for its safety.	3
		TOTAL	42

B.TECH II YEAR III SEMESTER FOOD TECHNOLOGY COURSE

SUBJECT CODE:**KFT 302**	COURSE TITLE: **FOOD MICROBIOLOGY**
EXAMINATION DURATION: **3 Hrs.**	SEMESTER: **III (ODD)**
L:3 T:0 P:0 CREDITS:04	**PRE-REQUISITE:** NIL

OBJECTIVE: To give the students an understanding of the role of microorganisms in food processing and preservation; the relation of microorganisms to food spoilage, foodborne illness, and intoxication; general food processing and quality control and the role of microorganisms in health promotion.

COURSE OUTCOME:

After successful completion of the course the students will be able to:

- Understand how microorganisms enter and grow in foods and Understand the growth requirements of common food borne pathogens and spoilage microorganisms.
- Understand intrinsic and environmental aspects of foods and how they can be modified to affect the growth and metabolism of food borne microorganisms to include spoilage, pathogenic, and beneficial organisms.
- Be able to attribute the characteristics of a food borne illness with unknown etiology to the most likely organism.
- To identify and control foodborne diseases and food spoilage

REFERENCE BOOKS:

S. No	Name of Authors / Books / Publishers	Year of Publication/ Reprint
1.	Banawart GJ, " Basic Food Microbiology" 2nd Edition, AVI Publishing Company	1989
2.	Frazier J & Westhoff DC, " Food Microbiology", 4th Edition, Tata McGraw Hill	2010
3.	Garbutt J," Essentials of Food Microbiology" , 2nd Edition, Hodder Arnold	1997
4.	Jay JM, Loessner MJ & Golden DA, " Modern Food Microbiology", 7th edition , Springer, NY	2005
5.	Ray B., " Fundamentals of Food Microbiology", 5th Edition, CRC Press	2013

COURSE DETAILS:

Units	S. No.	Contents	Lecture Hours
I	1.	General characteristics of microorganism: Classification, morphology, physiology, growth, nutrition and reproduction	4
	2	Pure culture techniques and maintenance of cultures	2
	3	Control of microorganisms.	2

II	**1**	Incidence of microorganism in foods, Sources of contamination	**3**
	2	Principles underlying spoilage of foods. Physical and Chemical methods to control microorganisms.	**3**
	3	Food poisoning and food borne infections, Microbial toxins, Indicator organisms.	**3**
III	**1**	Contamination, spoilage and preservation of cereal products & sugar products	**4**
	2	Contamination, spoilage and preservation of fruit and vegetable products, Bakery Products, Microbiological standard of foods.	**4**
IV	**1**	Contamination, spoilage and preservation of Meat products, Fish and Sea foods, Egg and Poultry products,	**4**
	2	Contamination, spoilage and preservation of Milk and Milk products and other foods. Microbiolgical limits.	**4**
V	**1**	Food plant sanitation, inspection and control, personnel hygiene	**3**
	2	Microbes in food fermentation, putrefaction, Lipolysis. Antagonism and Synergism in microorganisms.	**3**
	3	Rapid methods in detection of microorganisms. Standard plate count; Yeast and mould count	**3**
		TOTAL	**42**

SUBJECT CODE:**KFT 351**	COURSE TITLE: **ANALYSIS OF FOOD CONSTITUENTS LAB**
EXAMINATION DURATION: **2Hrs.**	SEMESTER: **III (ODD)**
L:0 T:0 P:2 CREDITS:01	**PRE-REQUISITE:** NIL

OBJECTIVE: To study the application and analytical procedures for characterizing the properties of foods and their constituents. These analytical procedures are used to provide information about a wide variety of different characteristics of foods, including their composition, structure, physicochemical properties and sensory attributes.

COURSE OUTCOME:

After successful completion of the course the students will be able to:

- Gain knowledge of practices for evaluation of appropriate methods for analysis of foods
- Understand various methodologies for analysis of components in foods.
- Knowledge of procedures for sampling, sample storage, preparation, Good Laboratory Practices, and proper record keeping
- Describe common instrumentation used in food analysis

LIST OF EXPERIMENTS:

1.Sampling requirements, procedures and methods.
2. Determination of moisture content of foods by oven drying and distillation methods.
3. Determination of Total and Acid insoluble ash content in foods.
4. Determination of Crude fat content by solvent extraction methods in foods.
5. Determination of crude Protein by Kjeldhal Lowry methods.
6. Determination of reducing and total sugar content in foods.
7. Determination of crude Fibre content in foods.
8. Determination of specific mineral contents in foods such as Ca, Iron, P, Chloride etc.
9. Determination of specific vitamin content of food such as ascorbic acid, carotenes etc.
10. Chromatographic Separation and identification of sugars and amino acids.

SUBJECT CODE:**KFT 352**	COURSE TITLE: **FOOD MICROBIOLOGY LAB**
EXAMINATION DURATION: **2Hrs.**	SEMESTER: **III (ODD)**
L:0 T:0 P:2 CREDITS:01	**PRE-REQUISITE:** NIL

OBJECTIVE: This is aimed to provide a hands-on opportunity for the students to practice and observe the principles of food microbiology. Students will familiarize themselves with the techniques used to research, regulate, prevent, and control the microorganisms found in food and understand the function of beneficial microorganisms during the food manufacturing process.

COURSE OUTCOME:

After successful completion of the course the students will be able to:

- Obtain a good understanding of the microscope , microbial stains and culture media used in microbiology.
- Obtain knowledge on how different processing, storage and environmental conditions affect microbial changes in food systems and how to adjust these conditions to improve or minimize microbial deterioration of foods systems

LIST OF EXPERIMENTS:

1. Microscope its parts and utility in identification and differentiation of various microorganisms as bacteria, yeast and mold.
2. Familiarization with common techniques for handling pure culture serial dilution, Inoculation, slide preparation incubation,counting etc.
3. Micrometry and determination of size of different microbes.
4. Simple and differential staining of microorganisms and their examination.
5. Direct total, viable, and non-viable count of microorganisms in milk.
6. Preparation and sterilization of media and glass ware for microbial counts.
7. Determination of Standard Plate Count (SPC) in natural and/or processed foods.
8. Microbiological examination of some selected natural and processed foods..
9. Microbiological examination of potable water: Total and coliform count.
10. Enumeration of coliform organism in some selected processed foods

SUBJECT CODE:**KFT 353**	COURSE TITLE: **FLUID FLOW & SOLID HANDLING LAB**
EXAMINATION DURATION: **2Hrs.**	SEMESTER: **III (ODD)**
L:0 T:0 P:2 CREDITS:01	**PRE-REQUISITE:** NIL

OBJECTIVE: To provide practice in estimating friction losses , impart training to use various flow measuring devices and to determine the various parameters related to fluid flow in pipes and in open channels.

COURSE OUTCOME:

After successful completion of the course the students will be able to:

- Estimate the friction and measure the frictional losses in fluid flow.
- Understand the functioning of various flow measurement devices as venturimeter & orifice meter
- Predict the coefficient of discharge for flow through pipes.

- Practically understand the various Unit Operations involving Solid Handling.

LIST OF EXPERIMENTS:

1. Study and verification of the flow pattern in a Bernoulli's apparatus.
2. To differentiate between laminar and turbulent flow using Reynolds experiment
3. To calibrate & determine the coefficient of discharge of an orifice meter.
4. To calibrate & determine the coefficient of discharge of an venturi meter.
5. Measurement of viscosity of Newtonian and non-Newtonian fluid.
6. To carry out differential and cumulative screen analysis of given sample of solid particles
7. To study the performance of Jaw Crusher/Ball mill.
8. Study of Pipe fittings and Valves.
9.To study the filtration characteristics of Rotary Vacuum /Plate and Frame filter.
10.To study the variation of friction factor, 'f' for turbulent flow in smooth and rough commercial pipes.

B.TECH II YEAR IV SEMESTER FOOD TECHNOLOGY COURSE

SUBJECT CODE:**KFT 401**	COURSE TITLE: **HEAT & MASS TRANSFER**
EXAMINATION DURATION: **3 Hrs.**	SEMESTER: **IV (EVEN)**
L:3 T:0 P:0 CREDITS:03	**PRE-REQUISITE:** NIL

OBJECTIVE: This course is designed to introduce a basic study of the phenomena of heat and mass transfer, to develop methodologies for solving a wide variety of practical engineering problems, and to provide useful information concerning the performance and design of particular systems and processes used in food processing operations.

COURSE OUTCOME:

After successful completion of the course the students will be able to:

- Understand the basic laws of heat transfer.
- Understand the fundamentals of conductive, convective and radiation heat transfer processes and evaluate Heat Transfer Coefficients for the same.
- Understand the Heat Exchangers and analyze heat exchanger performance.
- Understand the various mass transfer operations occurring during Diffusion and Drying etc.

REFERENCE BOOKS:

S. No	Name of Authors / Books / Publishers	Year of Publicatin/ Reprint
1.	F. Kreith, R.M. Manglik, M.S. Bohn," Principles of Heat Transfer", 7th Edition, Cengage Learning	2011
2.	Warren McCabe, Smith Julian, Peter Harriott," Unit Operations in Chemical Engineering",7th Edition, Tata McGraw Hill	2005
3.	Holman, JP, "Heat Transfer" , 9th edition , Tata McGraw Hill	2008
4.	Treybal, R.E., "Mass Transfer Operations", 3rd edition, Tata McGraw Hill	1981

COURSE DETAILS:

Units	S. No.	Contents	Lecture Hours
I	1	Conduction and Convection: Introduction, Conduction: Basic concepts of conduction in solids, liquids and gases	2
	2	Fourier law of heat conduction. Material properties of importance in heat transfer: Thermal conductivity, Specific heat capacity.	3
	3	General equation of heat conduction in cartesian co-ordinate, heat conduction through plane and composite walls, cylinders and spheres Critical and optimum insulation thickness.. Principles of convection, Newton's law of cooling.	4
II	1	Radiation: Basic laws of heat transfer by radiation, Absorptivity, reflectivity and transmissivity, black, white and grey body, emissive power and emissivity, laws of radiation – Planck, Stefan-Boltzmann, Wien's displacement, Kirchhoff's law.	4
	2	Introduction to Heat Transfer Equipments: Classification of heat exchangers, Double pipe, Shell & tube and Plate type heat exchanger, Evaporators.	4
III	1	Diffusion: Classification of mass transfer operation, choice of separation method, Definition of molecular and eddy diffusion,	3
	2	Ficks first law of diffusion, Concept of N & J Flux, Diffusion coefficient, Dependence of diffusion coefficient on temperature, pressure and composition.	3
	3	Mass Transfer Coefficients: Mass transfer in laminar and turbulent regions, F and k type mass transfer coefficients, Film, Penetration and surface renewal theories.	3
IV	1	Drying: Drying, Different modes of drying operation, Types of batch and continuous dryer	4
	2	Definitions of moisture contents, Rate of batch drying, Time of drying, Continuous drying.	4

V	**1**	Crystallization: Crystallization, Crystallization equipment, Classification of crystallizers	**4**
	2	Theories of crystallization, Factors governing nucleation and crystal growth rates, controlled growth of crystal.	**4**
		TOTAL	**42**

SUBJECT CODE:**KFT 402**	COURSE TITLE: **FOOD CHEMISTRY**
EXAMINATION DURATION: **3 Hrs.**	SEMESTER: **IV (EVEN)**
L:3 T:1 P:0 CREDITS:04	**PRE-REQUISITE:** NIL

OBJECTIVE: To understand the chemistry of foods - composition of food, role of each component and their interactions. The course is intended to give students an overview of the chemical and physical properties of the major and minor food components and their changes during processing, handling and storage.

COURSE OUTCOME:

After successful completion of the course the students will be able to:

- Have knowledge of the properties and structure of water molecules and nature of interaction with food macromolecules.
- Have knowledge of the structure of amino acids & assimilation to form primary, secondary, tertiary and quaternary structure of food proteins, functional properties of proteins and effect of processing on them
- Have Knowledge of structure of fatty acids and physical and chemical properties, effect of deep fat frying and modified fats
- Have knowledge of structure and isomerization of monosaccharides and disaccharides; starch, cellulose, pectin, alginate, gums and their chemical modifications and properties of use in foods.
- Have knowledge of vitamins, minerals, pigments and food additives. Usage of them in processed foods

REFERENCE BOOKS:

S. No	**Name of Authors / Books / Publishers**	**Year of Publication / Reprint**
1.	Fennema OR, " Food Chemistry", 5th Edition, CRC Press	2017

2.	Meyer LH, " Food Chemistry" CBS Publishers & Distributors	2004
3.	DeMan JM, " Principles of Food Chemistry" 3rd Edition, Aspen Publishers, Gaithersburg	1999
4.	H.-D. Belitz, W. Grosch," Food Chemistry", 4th Edition, Springer Science & Business Media	2009

COURSE DETAILS:

Units	S. No.	Contents	Lecture Hours
I	1	Water in Foods: Structure. Properties, Interactions, Water activity, Sorption Isotherms and food stability.	4
	2	Carbohydrates: Functions, Reactions and properties of simple and complex carbohydrate, Browning reactions, Selection of Natural or Modified carbohydrates for incorporation into processed food.	5
II	1	Lipids: Consistency of commercial fats, Lipolysis, Auto oxidation, Thermal Decomposition and effect of ionizing radiation,	4
	2	Refining of oils, Modification of oils and fats, Role of food lipids in flavor, Nutritional aspects of natural and modified fats.	4
III	1	Proteins: Structure, Chemical reactions and interactions of amino acids and proteins	4
	2	De-naturation and it implications, Functional properties of food proteins	3
	3	Modification of food proteins in processing and storage and its implications. Nutritive value of food proteins	2
IV	1	Vitamins, Minerals, Pigments and Flavours: Chemistry and stability of water and fat soluble vitamins, Chemical properties of minerals and their bioavailability, Enrichment and Fortification.	4
	2	Natural pigments in foods and their retention in processed foods. Flavoring constituents in foods, Development of process and reaction flavour volatiles.	4
V	1	Food Additives: Definitions, sources, uses and functions and regulatory aspects of Food additives. Limits of usage as per FSSAI.	5
	2	Intentional & Non Intentional additives, Food Adulterants.	3
		TOTAL	42

SUBJECT CODE:**KFT 403**	COURSE TITLE: **FOOD BIOCHEMISTRY**
EXAMINATION DURATION: **3**	SEMESTER: **IV (EVEN)**

Hrs.	
L:3 T:1 P:0 CREDITS:04	**PRE-REQUISITE:** NIL

OBJECTIVE: To furnish a student with knowledge and understanding of the basic biological and chemical processes of living cells and enzymes and how these are harnessed into industrial processes and technologies for the production, processing and preservation of food and related products.

COURSE OUTCOME:

After successful completion of the course the students will be able to:

- Understand the fundamental biochemical principles, structure/function of bio-molecules, metabolic pathways and the regulation of biochemical processes.
- Understand the the mechanism of enzyme action and identify factors affecting their action.
- Understand and be able to control the major chemical and biochemical (enzymatic) reactions that influence food quality with emphasis on food industry applications.
- Understand how the properties of different food components and interactions among these components modulate the specific quality attributes of food systems.

REFERENCE BOOKS:

S. No	Name of Authors / Books / Publishers	Year of Publication/ Reprint
1.	AL Lehninger, " Principle of Biochemistry", 7th Edition, W H Freeman & Co.	2017
2.	Lubert Stryen," Biochemistry", 6th Edition, W H Freeman & Co.	2006
3.	Shubhangini A. Joshi' "Nutrition and Dietetics" 2nd Edition Tata Mc GrawHill publishing Company Ltd.	2002
4.	JH Weil, " General Biochemistry", 6th Edition, New Age International Private Limited	1990

COURSE DETAILS:

Units	S. No.	Contents	Lecture Hours
I	1	Nomenclature, Classification and specificity of enzymes and cofactors	4
	2	Enzyme Kinetics: Factors affecting the rate of enzyme catalyzed reaction, regulation and control of enzyme action.	4
II	1	Metabolic Pathways: Carbohydrates, Proteins and Fats	5

	2	Catabolism and Anabolism	**3**
III	**1**	Digestion, Absorption, Assimilation and Transport of nutrients in human beings	**8**
IV	**1**	Post harvest and Postmortem biochemical changes in foods	**5**
	2	Changes in composition, color, texture, flavor and its implications on quality of foods.	**4**
V	**1**	Application of enzymes in food processing: Endogenous enzymes and their role in modification of foods,	**5**
	2	Enzymes added to foods during processing sources, conversions and specific applications.	**4**
		TOTAL	**42**

SUBJECT CODE:**KFT 451**	COURSE TITLE: **HEAT & MASS TRANSFER LAB**
EXAMINATION DURATION: **2 Hrs.**	SEMESTER: **IV (EVEN)**
L:0 T:0 P:2 CREDITS:01	**PRE-REQUISITE:** NIL

OBJECTIVE: To study and verift the various modes of Heat Transfer and to work on the Heat Transfer Equipments along with a practical study of important Mass Transfer Operations relevant to the industry.

COURSE OUTCOME:

After successful completion of the course the students will be able to:

- Apply principles of heat and mass transfer to basic engineering systems
- Analyse heat transfer by conduction, convection and radiation
- Analyse and design heat exchangers;
- Analyse diffusional processes and calculate the flux in a diffusion process.

LIST OF EXPERIMENTS:

1. To find out the thermal conductivities of Metal rod.
2. To find the thermal conductivity of Composite cylinder.
3. To find out the thermal conductivity of insulating powder.
4. To find the thermal conductivity of liquid / gases.
5. To determine the overall heat transfer coefficient in Parallel flow heat exchanger
6. To determine the overall heat transfer coefficient in Counter flow heat exchanger
7. To determine the overall heat transfer coefficient in Shell and Tube heat exchanger
8. Any Experiment on natural convection
9. Determination of diffusivity of acetone in air.
10. To determine the drying characteristics of given sample.

SUBJECT CODE:**KFT 452**	COURSE TITLE: **FOOD CHEMISTRY LAB**
EXAMINATION DURATION: **2Hrs.**	SEMESTER: **IV (EVEN)**
L:0 T:0 P:2 CREDITS:01	**PRE-REQUISITE:** NIL

OBJECTIVE: To study the analytical procedures for characterizing the properties of foods constituents and their interactions that affect the quality and stability of foods.

COURSE OUTCOME:

After successful completion of the course the students will be able to:

- Use the laboratory techniques common to basic and applied food chemistry.
- Understand the principles behind analytical technique when presented with a practical problem.
- Demonstrate practical proficiency in a food analysis laboratory.

LIST OF EXPERIMENTS:

1. Analysis of water for potable and food purposes
2. Moisture content in foods in relation to their stability
3. Non-enzymatic browning reactions and its determination
4. Determination of rate/ extent of hydrolysis of sucrose/starch

5. Determination of free fatty acid content in fats and oils
6. Detection and estimation of oxidative rancidity in fats/oils
7. Determination of heat stability of vitamin C
8. Study of some reactions of proteins
9. Study of some processing changes in proteins
10. Study of some functional properties of proteins

SUBJECT CODE:**KFT 453**	COURSE TITLE: **FOOD BIOCHEMISTRY LAB**
EXAMINATION DURATION: **2Hrs.**	SEMESTER: **IV (EVEN)**
L:0 T:0 P:2 CREDITS:01	**PRE-REQUISITE:** NIL

OBJECTIVE: To give the students a solid foundation in biology and chemistry and to develop analytical and critical thinking skills in biological phenomena through scientific methods.

COURSE OUTCOME:

After successful completion of the course the students will be able to:

- Learn fundamental approaches for experimentally investigating biochemical problems,
- Learn the enzyme kinetics
- Understand the applicability of the biochemical methods to realistic situations

LIST OF EXPERIMENTS:

1. Determination of enzyme activity and specific activity (Enzyme assay).
2. Determination of effect of temperature on enzyme activity.
3. Determination of effect of pH on enzyme activity.
4. Determination of effect of substrate concentration on enzyme activity & estimation of Km.
5. Estimation of enzymatic browning in foods.
6. Estimation of enhancement in an enzyme activity during ripening of fruits.
7. Estimation of enhancement in an enzyme activity during sprouting of grains.
8. Detection/ estimation of catalase and peroxidase activity in vegetable.

9. Application of enzymes in various foods.